LOCALE

Local **O**racle for **C**ausal **A**lignment via **L**earnable **E**go-graphs

Aaron Feng · Zhongyan Luo · DSC 190 · Prof. Biwei Huang

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Recap: The Landscape We Presented

Three paradigms for LLM-assisted causal discovery:

Paradigm	Example	Context/Query	Queries
Per-edge	MosaCD, chatPC	~200 tokens (one edge)	O(E × K)
Ego-graph	LOCALE (ours)	~500-1000 tokens (neighborhood)	O(N × K)
Whole-graph	CausalFusion	$O(V^2)$ tokens (full DAG)	O(rounds)

MosaCD: shuffled queries + non-collider-first propagation. Best F1 in 9/10 benchmarks.

But each query uses the LLM as a **binary classifier**.
128K context window. ~200 tokens used.

The Gap We Identified

Per-edge (MosaCD): one query = one binary vote. No local consistency.
CI constraints checked after the LLM, not during.

Whole-graph (CausalFusion): entire DAG in context. $O(V^2)$ tokens.
LLM does what Meek rules already do perfectly.

The missing middle: show the LLM one node's neighborhood.
It scores all incident edges **jointly**, sees CI constraints,
reasons about local mechanism consistency.

> The LLM's comparative advantage is contextual reasoning about local causal mechanisms — not binary edge classification, not global graph editing.

LOCALE Pipeline

Phase 0: PC-stable skeleton + separating sets (standard, no LLM)

Phase 1: Ego-graph LLM scoring
One query per node, K=10 stochastic votes with shuffled neighbor order.
LLM sees: node + all neighbors + CI constraints + cross-neighbor evidence.

Phase 2: CI-derived constraints via Max-2SAT (NCO variant)
Non-collider constraints are hard. LLM votes are soft.
Exact enumeration over feasible assignments per node.

Phase 3: Confidence-weighted dual-endpoint reconciliation
Each edge scored from both endpoints. Higher confidence wins.

Phase 4: Safety valve
Compare Phase 2 accuracy to Phase 1 per node. Revert where constraints hurt.
Ensures the pipeline is **monotonically non-decreasing**.

LLM does local reasoning. Algorithm handles global consistency.

The Ego-Graph Prompt

```
CENTER NODE: DrivingSkill — "Driver's overall skill level"

NEIGHBORS (4 variables):
  1. Age — "Driver's age group"
  2. SeniorTrain — "Senior training completed"
  3. DrivQuality — "Quality of driving"
  4. DrivHist — "Driving violation history"

STATISTICAL CONSTRAINTS:
  R1. Age ⊥ DrivQuality | {DrivingSkill} — NON-COLLIDER.
      Age and DrivQuality must NOT both point into DrivingSkill.
  R2. Age ⊥ DrivHist | {DrivingSkill} — NON-COLLIDER.
  R3. SeniorTrain ⊥ DrivQuality | {DrivingSkill} — NON-COLLIDER.

TASK: Orient each edge. Use constraints, then domain knowledge.
```

One query → 4 edge orientations. MosaCD: 4 × 10 = 40 queries.

The LLM sees constraints and domain knowledge **simultaneously** (phi × psi).

Finding 1: Ego Needs Scale

Ego-graph prompting requires sufficient model capacity.
Same edges, same skeleton, same CI evidence — only model size changes.

Model	Per-Edge	Ego-Graph	Delta
Qwen3.5-**4B**	90.5%	47.4%	**-43.2pp**
Qwen3.5-**9B**	90.5%	80.0%	-10.5pp
Qwen3.5-**27B**	93.7%	94.7%	**+1.1pp**

Joint neighborhood reasoning is a harder task than binary classification.
Small models fail catastrophically. At **27B**, ego matches or wins.

Insurance network, disguised variable names, K=5 votes. XN-001/002/003.

Finding 2: Skeleton Is the Bottleneck

Decomposing directed-edge F1: **skeleton coverage × orientation accuracy**

Network	Skeleton Coverage	Orientation Acc.	Directed F1	MosaCD F1
Insurance	83%	**100%**	88.4%	87%
Asia	88%	**100%**	93.3%	93%
Alarm	88%	89.9%	89.9%	93%
Child	88%	88.0%	88.0%	90%
Hepar2	**52%**	80.6%	58.8%	72%

On edges the skeleton gets right, orientation is **at ceiling**.
The F1 gap vs MosaCD is skeleton coverage, not orientation quality.

Hepar2: 52% skeleton coverage. No orientation method can fix that.

Finding 3: The NCO Discovery

Phase 2 (hard CI constraints) **hurt** on 3/6 networks. Why?

~98% of incorrect CI-derived orientation facts are false colliders.

Network	CI Accuracy	False Colliders	False Non-Colliders	P2 Impact
Insurance	92.4%	7	**0**	+4.2pp
Alarm	90.4%	7	**0**	+1.3pp
Child	78.9%	12	**0**	+6.2pp
Sachs	95.5%	1	**0**	-5.9pp
Asia	90.0%	1	**0**	-16.7pp
Hepar2	65.1%	52	**0**	-9.9pp

Non-collider = positive evidence (Z IS in sep set). Always reliable.
Collider = absence of evidence (Z NOT FOUND). Fails with finite samples.

NCO: The Fix and Its Generality

Fix: use only non-collider constraints. 100% reliable.

Network	Ego Only	+ Hard Constraints	+ NCO Only
Insurance	93.1%	97.2%	**98.6%**
Alarm	88.3%	89.6%	89.6%
Sachs	70.6%	64.7%	**70.6%**
Asia	100%	83.3%	**100%**
Hepar2	75.7%	65.8%	**79.3%**
Aggregate	83.7%	81.1%	**86.7%**

NCO never hurts. Hard constraints hurt 3/6 networks.

This is **method-agnostic**: applies to MosaCD, PC, any CI-based method.
Validated at n=1k through n=10k. 100% false-collider rate at n ≥ 5000.

Finding 4: Head-to-Head vs MosaCD (Same Model)

Fair comparison: **same Qwen3.5-27B, same skeleton, same data**.
MosaCD reimplemented with identical LLM backend.

Network	LOCALE	MosaCD	Result
Insurance	**86.3%**	86.3%	Tie (LOCALE **46% fewer queries**)
Alarm	**89.9%**	80.9%	**LOCALE +9.0pp**
Child	**88.0%**	88.0%	Tie (LOCALE **70% fewer queries**)
Asia	**93.3%**	93.3%	Tie (LOCALE **29% fewer queries**)
Sachs	**76.5%**	58.8%	**LOCALE +17.7pp**

Score: 2 wins, 3 ties, 0 losses. Always ~50% fewer queries.
On ties, LOCALE matches MosaCD with substantially fewer LLM calls.

Finding 5: Query Efficiency

Ego-graph batching: one query scores **all** incident edges.

Per-edge: one query per edge × K votes = O(E × K) queries
Ego-graph: one query per node × K votes = O(N × K) queries

Across 6 networks: **2.4-4.0x fewer queries** (58-75% savings)

Cost scales with **nodes**, not edges.
As graphs get denser, the savings grow.

At 27B, each ego query costs more tokens but produces
d edges of information instead of 1.
Net result: competitive accuracy at a fraction of the cost.

Sachs (11 nodes, 17 edges): 2.4x. Insurance (27 nodes, 52 edges): 3.8x.

Ablation: What Each Phase Contributes

Full pipeline: **+6.6pp** over per-edge baseline (85.6% → 92.2%)

Phase	What It Does	Cumulative	Delta
PE baseline	Per-edge LLM votes	85.6%	—
+ P1 Ego scoring	Joint neighborhood context	87.0%	+1.4pp
+ P2 NCO constraints	CI-feasible assignments	88.8%	+1.9pp
+ P3 Reconciliation	Dual-endpoint agreement	89.7%	+0.8pp
+ P4 Safety valve	Damage detection + revert	**92.2%**	**+2.5pp**

Safety valve is the largest single contributor.

Not cleanup — core design. Detects when constraints hurt and reverts.

4 networks, 215 edges, orientation accuracy on skeleton edges. XN-012.

What Didn't Work

Dawid-Skene reconciliation (proposed)
Underperforms majority vote. Only 2 annotators per edge — too sparse for EM.

Meek propagation (proposed)
Complete no-op. Max-2SAT already orients everything. Nothing left for Meek.

Skeleton refinement via LLM
0 true positives across 6 networks. Most missing edges are >2 hops apart.
Hepar2 worst: 9 false positives, 0 true positives. Definitively vetoed.

Ego at degree 3 ("valley of confusion")
Too few CI constraints to guide, enough complexity to confuse.
Per-edge wins at d=3. Ego wins at d≥4.

Iterative BFS expansion (in progress)
SP-style decimation: established edges feed as context into later rounds.
Initial result: single-pass K=10 (F1=0.863) > multi-round K=5×2 (F1=0.800).
Root cause: error amplification from early decimation. More votes > more context.
Still exploring: matched budgets, different thresholds, incremental seeds.

What MosaCD Got Right, Where LOCALE Adds

MosaCD's insight confirmed: non-collider-first propagation works.
Our NCO finding provides the **theoretical grounding** — it's not just empirically better. We now know why: ~98% of CI errors are false colliders.

What LOCALE adds:

1. **NCO as a general principle** — method-agnostic, not MosaCD-specific
2. **F1 decomposition** — skeleton is the binding constraint (reframes the field)
3. **Same accuracy, half the queries** — 2 wins, 3 ties, 0 losses vs MosaCD (same model)
4. **Open-source viability** — 27B matches proprietary on most benchmarks
5. **Safety valve** — monotonically non-decreasing pipeline

Where both struggle: skeleton coverage.
Neither can overcome a bad PC skeleton. That's the real frontier.

Limitations

- **Causal sufficiency** assumed — no latent confounders
- **Single model family** — only Qwen3.5 tested (no GPT-4o, Claude, Llama)
- **PC skeleton ceiling** — LOCALE cannot recover edges PC misses
- **n=1000** for most experiments — CI tests degrade on large graphs
- **Calibration** (Venn-Abers) not yet validated — needs held-out graph instances
- **Iterative expansion** initial result negative — still exploring matched-budget variants

Contributions and What's Next

Five contributions:

1. **NCO discovery**: ~98% of CI errors are false colliders (method-agnostic)
2. **F1 decomposition**: skeleton, not orientation, is the binding constraint
3. **Same accuracy, half the queries**: 2 wins, 3 ties vs MosaCD on same model
4. **Open-source parity**: 27B matches proprietary models
5. **Safety valve**: monotonically non-decreasing pipeline design

What's next:

- Better skeletons: hybrid statistical-LLM skeleton learning (the real bottleneck)
- Cross-model validation: GPT-4o, Claude, Llama
- Calibrated selective output (Venn-Abers edge confidence)
- NCO as preprocessing for all CI-based methods

Thank You

Questions?

Code and experiments: projects/locale/ · Qwen3.5-27B on RunPod vLLM